Problem Set 2: Solutions

Econon 502: Advanced Microeconomics

Problem 1: Automation and Labor Demand

a) Elasticity of substitution

The CES production function is: $Q = [0.6R^{0.5} + 0.4L^{0.5}]^2$

This can be rewritten as: $Q = [\alpha R^\rho + \beta L^\rho]^{\gamma/\rho}$ where $\rho = 0.5$ and $\gamma = 2$.

The elasticity of substitution is:

$$\sigma = \frac{1}{1-\rho} = \frac{1}{1-0.5} = 2$$

This means robots and workers are relatively **easily substitutable**. A 1% change in the relative wage ratio leads to a 2% change in the robot-to-worker ratio. This is higher than Cobb-Douglas ($\sigma = 1$), indicating the firm can readily replace workers with robots when relative prices change.

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b) Cost-minimizing mix for Q = 1000

The cost minimization condition is:

$$\frac{MP_R}{MP_L} = \frac{r_R}{w}$$

First, find marginal products:

$$MP_R = \frac{\partial Q}{\partial R} = 2[0.6R^{0.5} + 0.4L^{0.5}] \cdot 0.6 \cdot 0.5R^{-0.5} = \frac{0.6R^{-0.5}}{[0.6R^{0.5} + 0.4L^{0.5}]^{-1}}$$

By symmetry (and using the CES tangency condition):

$$\frac{0.6R^{-0.5}}{0.4L^{-0.5}} = \frac{20}{25}$$

Solving:

$$\frac{0.6L^{0.5}}{0.4R^{0.5}} = 0.8$$

$$\frac{L^{0.5}}{R^{0.5}} = \frac{0.8 \times 0.4}{0.6} = 0.533$$

$$\frac{L}{R} = 0.284$$

So $L = 0.284R$.

Substitute into production function with $Q = 1000$:

$$1000 = [0.6R^{0.5} + 0.4(0.284R)^{0.5}]^2$$

$$31.62 = 0.6R^{0.5} + 0.4(0.533)R^{0.5}$$

$$31.62 = 0.813R^{0.5}$$

$$R^* = 1512, \quad L^* = 430$$

Cost: $C = 20(1512) + 25(430) = 30,240 + 10,750 = \$41,000$

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c) Robot rental rate falls to $r_R = 10$

New tangency condition:

$$\frac{0.6L^{0.5}}{0.4R^{0.5}} = \frac{10}{25} = 0.4$$

$$\frac{L}{R} = \left(\frac{0.4 \times 0.4}{0.6}\right)^2 = 0.071$$

So $L = 0.071R$.

For $Q = 1000$:

$$31.62 = 0.6R^{0.5} + 0.4(0.266)R^{0.5} = 0.706R^{0.5}$$

$$R^* = 2006, \quad L^* = 142$$

Robot-to-worker ratio change: - Before: $R/L = 1512/430 = 3.52$ - After: $R/L = 2006/142 = 14.13$ - **Percentage change:** $(14.13 - 3.52)/3.52 = 301\%$ increase

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d) Robot tax analysis

If the robot tax raises r_R back to 20, we return to the solution in part (b): $L = 430$ workers.

Jobs saved: $430 - 142 = 288$ jobs per 1000 units

Trade-off: - Cost with $r_R = 10$: $10(2006) + 25(142) = \$23,610$ - Cost with $r_R = 20$: $\$41,000$ (from part b) - **Extra cost:** $\$17,390$ per 1000 units, or $\$60$ per job saved

The robot tax protects jobs but increases production costs by 74%, which could make the firm less competitive.

Problem 2: Renewable Energy Transition

a) Cost-minimizing combination

Production function: $Q = 10C^{0.4}S^{0.6}$

Cost minimization:

$$\frac{MP_C}{MP_S} = \frac{p_C}{p_S}$$
$$\frac{4C^{-0.6}S^{0.6}}{6C^{0.4}S^{-0.4}} = \frac{30}{50}$$
$$\frac{4S}{6C} = 0.6$$
$$\frac{S}{C} = 0.9$$

So $S = 0.9C$.

Substitute into $Q = 1000$:

$$1000 = 10C^{0.4}(0.9C)^{0.6} = 10C^{0.4} \cdot 0.954C^{0.6} = 9.54C$$

$$C^* = 105, \quad S^* = 94$$

Cost: $30(105) + 50(94) = \$7,850$

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b) Carbon tax of $15

New coal price: $p_C = 45$

New tangency:

$$\frac{S}{C} = \frac{6C \times 45}{4S \times 50} = 1.35$$

So $S = 1.35C$.

For $Q = 1000$:

$$1000 = 10C^{0.4}(1.35C)^{0.6} = 10C \times 1.197 = 11.97C$$
$$C^* = 84, \quad S^* = 113$$

New coal-to-solar ratio: $C/S = 84/113 = 0.74$ (down from $105/94 = 1.12$)

Cost: $45(84) + 50(113) = \$9,430$

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c) Carbon tax for 50% coal reduction

Target: $C = 52.5$ (half of 105)

For $Q = 1000$: $S = 1000/(10 \times 52.5^{0.4}) = 155$

Tangency condition: $\frac{4\times155}{6\times52.5} = \frac{p_C}{50}$

Solving: $p_C = \$98$ per ton

Required carbon tax: $98 - 30 = \$68$ per ton

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d) CEO's concern

Cost increase: $\$9,430 - \$7,850 = \$1,580$ (20% increase)

The CEO has a point that costs increase, but "bankrupt" is an exaggeration. The firm remains profitable as long as revenue exceeds $\$9,430$ per 1000 MWh. If competitors face the same carbon tax, all prices will rise proportionally, protecting margins.

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e) Extension: Solar cost decline

Current: 75% solar means $S/(C+S) = 0.75$, so $S = 3C$.

Tangency: $\frac{4\times 3C}{6C} = \frac{30}{p_S}$

$$2 = \frac{30}{p_S} \implies p_S = \$15$$

Solar cost must fall from \$50 to \$15 (70% decline).

With 10% annual decline: $(1-0.1)^n = 0.3 \implies n = \frac{\ln(0.3)}{\ln(0.9)} = 11.4$ years

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Problem 3: Platform Economics - Uber's Surge Pricing

a) Marginal cost

$$MC = \frac{dC}{dQ} = 5 + 0.02Q$$

The quadratic term ($0.01Q^2$) represents **congestion costs** or **decreasing returns to scale**. As Uber adds more rides per hour, costs increase at an increasing rate—perhaps due to longer wait times, need to dispatch drivers farther away, or system strain.

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b) Normal hours monopoly pricing

Demand: $P = 25 - 0.05Q$
Revenue: $R = PQ = 25Q - 0.05Q^2$
Marginal revenue: $MR = 25 - 0.1Q$

Set $MR = MC$:

$$25 - 0.1Q = 5 + 0.02Q$$

$$20 = 0.12Q$$

$$Q^* = 167 \text{ rides}$$

Price: $P^* = 25 - 0.05(167) = \$16.65$

Lerner Index:

$$LI = \frac{P - MC}{P} = \frac{16.65 - (5 + 0.02 \times 167)}{16.65} = \frac{16.65 - 8.34}{16.65} = 0.50$$

This indicates substantial market power (50% markup over marginal cost).

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c) Surge pricing

Demand: $P = 40 - 0.05Q$
MR: $MR = 40 - 0.1Q$

Set $MR = MC$:

$$40 - 0.1Q = 5 + 0.02Q$$

$$Q^* = 292 \text{ rides}$$

Price: $P^* = 40 - 0.05(292) = \$25.40$

Surge multiplier: $25.40/16.65 = 1.53$ (53% increase)

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d) Economic validity of surge pricing justification

Marginal cost increases with quantity ($MC = 5 + 0.02Q$), so higher demand does require higher prices to "bring drivers onto the road."

At normal time: $MC = 5 + 0.02(167) = \$8.34$
At surge time: $MC = 5 + 0.02(292) = \$10.84$

The marginal cost is genuinely higher during surge times, validating Uber's claim that surge pricing is needed to cover the increased cost of providing more rides. However, the markup over MC (Lerner Index) remains constant at 0.50, showing Uber also extracts monopoly rents.

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e) Price cap at 1.5x normal price

Price cap: $P_{cap} = 1.5 \times 16.65 = \$24.98$

At surge time, quantity supplied where $P = MC$:

$$24.98 = 5 + 0.02Q_S$$

$$Q_S = 1000 \text{ rides}$$

But wait—this violates our monopoly model. Let me recalculate assuming Uber supplies where $P_{cap} = MC$:

Actually, Uber won't supply where $P = MC$ if it can restrict quantity. At the price cap, Uber sets $MR = MC$ given the cap is binding.

At $P = 24.98$: Demand gives $Q_D = (40 - 24.98)/0.05 = 300$ rides

Uber's profit-maximizing quantity at this price: Since $P_{cap} > MC$ for reasonable Q, Uber supplies to meet demand.

Shortage: Minimal, as $Q_D \approx Q^* = 292$. The price cap barely binds.

Problem 4: Gig Economy - DoorDash Driver Decisions

a) Economic wage

Economic wage = Gross wage - Expenses = \$25 - \$5 = **\$20/hour**

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b) Utility maximization

Budget constraint: $C = 20L$ (where L is hours worked)
Time constraint: Leisure $= 40 - L$

Utility: $U = \ln(C) + 2\ln(40 - L) = \ln(20L) + 2\ln(40 - L)$

First-order condition:

$$\frac{dU}{dL} = \frac{1}{L} - \frac{2}{40 - L} = 0$$

$$\frac{40 - L}{L} = 2$$

$$40 - L = 2L$$

$$L^* = 13.33 \text{ hours}$$

Consumption: $C^* = 20(13.33) = \$267$
Utility: $U^* = \ln(267) + 2\ln(26.67) = 5.59 + 6.58 = 12.17$

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c) Busy time bonus

For 10 hours at $w = 35$ (net \$\$30/hour): - Hours worked: 10 + some regular hours - Income from bonus: $30 \times 10 = \$300$

Optimize remaining 30 hours between regular work (at $\$20/hr)andleisure: U = \ln(300 + 20L_r) + 2\ln(30 - L_r)$\$

$$\frac{20}{300 + 20L_r} = \frac{2}{30 - L_r}$$

$$20(30 - L_r) = 2(300 + 20L_r)$$

$$600 - 20L_r = 600 + 40L_r$$

$$L_r = 0$$

So work only the 10 bonus hours.

New utility: $U = \ln(300) + 2\ln(30) = 5.70 + 6.80 = 12.50$

Change: $\Delta U = 12.50 - 12.17 = 0.33$ (definitely accept!)

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d) Gas price increase

New economic wage: $25 - 8 = \$17/hour$

New optimum:

$$\frac{40 - L}{L} = 2 \implies L^* = 13.33 \text{ hours}$$

Labor supply is unchanged! This is because the utility function has income and substitution effects that exactly offset. The driver maintains the same hours despite lower real wages, just consuming less.

Labor supply is vertical (neither upward nor downward sloping) for this driver.

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e) Minimum wage policy

Policy requires net wage $\geq \$20/hour$.

With gas at $$8/hour, gross wage must be $ $28/hour.

Winners: - Drivers who continue working earn more ($28 vs $25 gross) - They maintain same hours (13.33) but higher consumption

Losers: - Platforms may reduce number of drivers - Customers face higher prices - Some drivers may lose jobs if demand is elastic

Ambiguous: - Total driver welfare depends on employment effects

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Problem 5: Netflix vs. Movie Theaters - Quality Choice

a) Profit-maximizing quality

Theaters:

$$\pi_T = 100q - q^2 - (50 + 10q + q^2) = 90q - 2q^2 - 50$$
$$\frac{d\pi_T}{dq} = 90 - 4q = 0 \implies q_T^* = 22.5$$

Profit: $\pi_T^* = 90(22.5) - 2(22.5)^2 - 50 = 2025 - 1012.5 - 50 = \$962.5$ million

Streaming:

$$\pi_S = 60q - 0.5q^2 - (20 + 5q + 0.5q^2) = 55q - q^2 - 20$$
$$\frac{d\pi_S}{dq} = 55 - 2q = 0 \implies q_S^* = 27.5$$

Profit: $\pi_S^* = 55(27.5) - (27.5)^2 - 20 = 1512.5 - 756.25 - 20 = \$736.25$ million

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b) Optimal channel

Choose **theaters** (profit difference: $962.5 - 736.25 = \$226.25$ million)

Interestingly, streaming allows higher quality ($q_S = 27.5 > q_T = 22.5$) but lower profit.

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c) COVID-19 impact

New theater revenue: $R_T = 40q - q^2$

$$\pi_T = 40q - q^2 - 50 - 10q - q^2 = 30q - 2q^2 - 50$$
$$q_T^* = 7.5, \quad \pi_T^* = 30(7.5) - 2(56.25) - 50 = \$62.5 \text{ million}$$

Now choose **streaming** (\$736.25 million > \$62.5 million).

COVID fundamentally changed the economics, making streaming dominant.

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d) Netflix's \$30 million offer

Streaming profit with upfront payment: $736.25 + 30 = \$766.25$ million
Theater profit: \$962.5 million (pre-COVID)

Reject the offer pre-COVID, **accept** during COVID.

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e) 20% streaming cost reduction

New costs: $C_S(q) = 20 + 4q + 0.4q^2$

$$\pi_S = 60q - 0.5q^2 - 20 - 4q - 0.4q^2 = 56q - 0.9q^2 - 20$$
$$q_S^* = 31.1, \quad \pi_S^* = \$850 \text{ million}$$

Still less than theaters (\$962.5 million).

Indifference point: Set $\pi_S = \pi_T = 962.5$

This requires further cost reductions. As streaming costs continue falling, theatrical releases become increasingly uncompetitive, explaining the industry's shift to streaming-first strategies.

Problem 6: The College Textbook Market

a) Cost components

- $200,000$: Fixed costs (editing, typesetting, author advances, development)
- $40Q$: Constant marginal cost (printing, shipping per book)
- $0.01Q^2$: Increasing marginal costs (rush orders, quality control strain, inventory costs)

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b) Monopoly pricing

Demand: $P = 300 - 0.02Q$
Revenue: $R = 300Q - 0.02Q^2$

MR: $MR = 300 - 0.04Q$
MC: $MC = 40 + 0.02Q$

Set $MR = MC$:

$$300 - 0.04Q = 40 + 0.02Q$$
$$Q^* = 4,333 \text{ books}$$

Price: $P^* = 300 - 0.02(4333) = \$213.34$

Profit:

$$\pi = 213.34(4333) - 200,000 - 40(4333) - 0.01(4333)^2 = \$551,556$$

Lerner Index:

$$MC(4333) = 40 + 0.02(4333) = \$126.66$$
$$LI = \frac{213.34 - 126.66}{213.34} = 0.406$$

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c) Consumer surplus and deadweight loss

Perfect competition: $P = MC$

$$300 - 0.02Q = 40 + 0.02Q$$
$$Q_c = 6,500, \quad P_c = \$170$$

Consumer surplus (monopoly):

$$CS_M = \frac{1}{2}(300 - 213.34)(4333) = \$187,652$$

Consumer surplus (competition):

$$CS_C = \frac{1}{2}(300 - 170)(6500) = \$422,500$$

Deadweight loss:

$$DWL = \frac{1}{2}(213.34 - 170)(6500 - 4333) = \$46,973$$

Students lose $\$234,848$ in consumer surplus $(422,500 - 187,652)$, while publisher gains $\$551,556$ in profit.

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d) Used book market

New demand: $P = 300 - 0.04Q$ (more elastic)
MR: $MR = 300 - 0.08Q$

$$300 - 0.08Q = 40 + 0.02Q$$
$$Q^* = 2,600, \quad P^* = \$196$$

Who benefits: - **Students:** Lower price ($196 < $213), though fewer new books sold - **Publisher loses:** Lower quantity and price - **Society:** Smaller deadweight loss due to more elastic demand constraining market power

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e) Open-source textbooks

FOR: - Eliminates deadweight loss of $\$46,973$ - Increases access (6,500 vs 4,333 students) - Reduces student costs by $$213 per student

AGAINST: - Publisher loses $\$551,556$ in profit - May reduce incentives for quality textbook development - Authors may not write textbooks without royalties - Fixed costs ($\$200,000$) must be covered by someone (universities? taxpayers?)

Efficiency argument: Open-source eliminates deadweight loss and increases total surplus, but requires alternative funding mechanisms for textbook development.

Problem 7: Amazon's Warehouse Automation

a) Current cost-minimizing mix

Production: $Q = 20K^{0.5}L^{0.5}$

Tangency: $\frac{MP_K}{MP_L} = \frac{r}{w}$

$$\frac{10K^{-0.5}L^{0.5}}{10K^{0.5}L^{-0.5}} = \frac{8}{15}$$

$$\frac{L}{K} = \frac{8}{15} \implies L = 0.533K$$

For $Q = 10,000$:

$$10,000 = 20K^{0.5}(0.533K)^{0.5} = 20K \times 0.730 = 14.6K$$

$$K^* = 685 \text{ robots}, \quad L^* = 365 \text{ workers}$$

Cost: $8(685) + 15(365) = \$10,955/hour$

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b) Investment comparison

Option A: $Q = 25K^{0.5}L^{0.5}$

For $Q = 10,000$: $10,000 = 25K \times 0.730 = 18.25K$

$$K^* = 548, \quad L^* = 292$$

Cost: $8(548) + 15(292) = \$8,764/hour$
Savings: $10,955 - 8,764 = \$2,191/hour$

Option B: $Q = 20K^{0.5}L^{0.6}$

New tangency: $\frac{10K^{-0.5}L^{0.6}}{12K^{0.5}L^{-0.4}} = \frac{8}{15}$

$$\frac{10L}{12K} = 0.533 \implies L = 0.640K$$

For $Q = 10,000$: $10,000 = 20K^{0.5}(0.640K)^{0.6}$

$$K^* = 614, \quad L^* = 393$$

Cost: $8(614) + 15(393) = \$10,807/hour$
Savings: $10,955 - 10,807 = \$148/hour$

Choose Option A (better robots save $\$2,191/hour$ vs $\$148/hour$ for training)

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c) Minimum wage increase

New tangency with $w = 20$:

$$\frac{L}{K} = \frac{8}{20} = 0.4 \implies L = 0.4K$$

For $Q = 10,000$:

$$10,000 = 20K^{0.5}(0.4K)^{0.5} = 12.65K$$

$$K^* = 791, \quad L^* = 316$$

K/L ratio: - Before: $685/365 = 1.88$ - After: $791/316 = 2.50$ - **Change:** 33% increase in capital intensity

Employment: Falls from 365 to 316 workers (13% decline)

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d) Automation and jobs

Amazon's claim has merit but is incomplete:

Substitution effect: Higher automation → fewer workers (we see L falls from 365 to 316)

Output effect: Lower costs → can produce more at lower prices → higher demand → need more inputs

If demand is sufficiently elastic, total employment could increase even as the K/L ratio rises. However, for a *given* output level (10,000 packages), automation clearly reduces employment.

Verdict: True in the long run if Amazon expands, misleading in the short run for existing operations.

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e) Forecasting robot adoption

Target: $K/L = 2 \times 1.88 = 3.76$

Current ratio with $r = 8, w = 15$: $K/L = 1.88$

New ratio: $\frac{K}{L} = \frac{r_t}{w} = \frac{r_t}{15} = 3.76 \implies r_t = 4$

Robot cost must fall from \$8 to \$4 (50% decline).

With 5% annual decline: $(1-0.05)^n = 0.5$

$$n = \frac{\ln(0.5)}{\ln(0.95)} = 13.5 \text{ years}$$

Problem 8: Spotify's Freemium Model

a) Current profit

Revenue: $3(200M) + 10(100M) = \$600M + \$1,000M = \$1,600M$
Cost: $5M + 2(300M) = \$605M$

Profit: $\$995M$

Marginal cost: $MC = \$2$ per user

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b) Drop free service

If 100M free users convert to premium: - Revenue: $10(200M) = \$2,000M$ - Cost: $5M + 2(200M) = \$405M$ - **Profit:** $\$1,595M$

Yes, drop free service (profit increases by $\$600M$)

However, this ignores strategic considerations: free users may later convert, provide network effects, and generate data.

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c) Better recommendations investment

New cost: $605M + 500M = \$1,105M$
New revenue: $3(170M) + 10(130M) = \$510M + \$1,300M = \$1,810M$

New profit: $\$705M$

No, don't invest (profit falls from $\$995M$ to $\$705M$)

The $\$500M$ investment only generates $210M$ in additional revenue (\$7 per converted user × 30M users).

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d) Royalty increase

New MC = \$3 per user

Current structure profit: - Revenue: $\$1,600M$ (unchanged) - Cost: $5M + 3(300M) = \$905M$ - **Profit:** $\$695M$ (down from $\$995M$)

Options: 1. **Raise premium to \$\$11:** If demand is inelastic, keep ~100M users, revenue = $\$1,100M + \$600M = \$1,700M$, profit = $\$795M$ 2. **Cut free users:** Reduces costs most 3. **Both:** Optimize on each margin

Best strategy: Raise prices modestly and reduce free user perks (fewer skips, more ads) to encourage premium conversion.

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e) Apple Music competition

Apple at $$9/month, premium only: - Assume 100M users - MC = $2 (similar infrastructure) - Profit per user: $9 - 2 = \$7$

Apple's Lerner Index: $LI = \frac{9-2}{9} = 0.78$

Spotify premium at $10: - Profit per user: $10 - 2 = \$8$ - **Lerner Index:** $LI = \frac{10-2}{10} = 0.80$

Winner: Depends on user acquisition: - **Spotify** if freemium converts users (uses free tier as marketing) - **Apple** if premium-only attracts quality-seeking users who dislike ads

Spotify's advantage: larger user base (300M total) and data from free users for better recommendations. Apple's advantage: simpler model, integrated ecosystem, no ad infrastructure needed.

In practice, both coexist by targeting different user segments.